

Question:

In a dance class of 20 students, 20% enrolled in contemporary dance, 25% of the remaining enrolled in jazz dance, and the rest enrolled in hip-hop dance. What percentage of the entire students enrolled in hip-hop dance?

Reasoning + Answer:

<think> To solve this problem , let's first determine the number of students who enrolled in contemporary dance . Given that there are 20 students in total and 20 % of them enrolled in contemporary dance , we can calculate this number as follows : $(20 \times 0.20 = 4)$ students enrolled in contemporary dance .

Next , we need to find out how many students are left after those enrolled in contemporary dance . We subtract the 4 students from the total of 20 students : $(20 - 4 = 16)$ students remaining .

Now , we focus on the students enrolled in jazz dance . The problem states that 25 % of the remaining students (after contemporary dance) enrolled in jazz dance . So , we need to calculate 25 % of the 16 remaining students : $(16 \times 0.25 = 4)$. Therefore , 4 students enrolled in jazz dance .

Finally , to find out how many students enrolled in hip-hop dance , we need to subtract the number of students enrolled in contemporary and jazz dance from the total number of students . We know there are 20 students in total , 4 in contemporary and 4 in jazz dance . So , we subtract these numbers from the total : $(20 - 4 - 4 = 12)$.

Now we have determined that 12 students enrolled in hip-hop dance . To find out what percentage this number is of the entire student body , we use the formula : $(\frac{ 12 }{ 20 } \times 100)$. Simplifying this , we get $(\frac{ 12 \times 100 }{ 20 } = \frac{ 1200 }{ 20 } = 60)$. </think>

<answer> 60 </answer>

